

Geometric Renormalization of the Speed of Light and the Origin of the Planck Scale in a Saturation-Stitch Vacuum

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Abstract

Attempts to reconcile General Relativity with Quantum Mechanics often falter on the problem of discretizing spacetime without violating Lorentz invariance. In this work, we propose the Selection-Stitch Model (SSM), a discrete vacuum framework based on a saturated Face-Centered Cubic (FCC) lattice ($K = 12$). We demonstrate two fundamental geometric renormalizations. First, we show that the physical speed of light (c) emerges as a renormalization of the lattice hopping speed, specifically $c = 4v_{\text{lattice}}$, due to the constructive interference of the 12 nearest-neighbor paths on the Cuboctahedron unit cell. Second, utilizing this relation, we derive the lattice spacing a not as a free parameter, but as the inevitable geometric limit where the vacuum's elastic energy density strikes the absolute $1/\sqrt{3}L$ kinematic exclusion limit (the metric wall). Fixed by the packing efficiency of the FCC unit cell ($V = a^3/\sqrt{2}$), we derive a fundamental lattice spacing of $a \approx 0.77l_P$. We explicitly derive the continuum limit, showing that Lorentz invariance is restored for fermions up to $\mathcal{O}(a^2)$ corrections due to the centrosymmetry of the lattice. Furthermore, we demonstrate that this same lattice centrosymmetry enforces the vanishing of the QCD topological charge density, naturally resolving the Strong CP problem ($\theta_{QCD} = 0$) without requiring an axion. Finally, we discuss the geometric obstruction to point-particle quantization of gravity in this framework.

1 Introduction

The unification of Quantum Field Theory (QFT) and General Relativity remains the paramount challenge of modern physics. A central obstacle is the continuous nature of spacetime assumed in QFT, which leads to ultraviolet divergences when applied to gravity. Discrete spacetime models offer a natural regularization mechanism [1], but they typically suffer from the breakdown of Lorentz invariance (anisotropy) at the grid scale [2].

We present the Selection-Stitch Model (SSM), which models the vacuum as an emergent, discrete, and geometrically saturated network. Unlike hypercubic models, the SSM utilizes a Face-Centered Cubic (FCC) lattice ($K = 12$), the densest possible sphere packing [4], which possesses sufficient symmetry to recover isotropy in the continuum limit.

Recent computational verifications of this network’s kinematics [3] have proven the existence of a strict geometric exclusion limit—a “metric wall” at exactly $1/\sqrt{3}L$ —beyond which vacuum nodes physically cannot be compressed.

In this paper, we derive:

1. The Geometric Renormalization of Light, deriving $c = 4v_{lattice}$.
2. The Origin of the Planck Scale, deriving $a \approx 0.77l_P$ from the geometry of the FCC unit cell and the $1/\sqrt{3}L$ metric wall.
3. The Restoration of Lorentz Invariance for fermions.
4. The Resolution of the Strong CP Problem via lattice centrosymmetry.

2 Geometric Emergence of Scales

2.1 Geometric Boost of Light Speed ($c = 4$)

The propagation of a fermion $\psi(x)$ is governed by a discrete hopping operator summed over the 12 nearest-neighbor vectors $\hat{n}_j$ of the cuboctahedral unit cell. We define the lattice vectors normalized by the spacing a :

$$n \in \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\} \quad (1)$$

The discrete Dirac operator D_{SSM} is:

$$D_{SSM}(k) = \frac{1}{\tau} \sum_{j=1}^{12} (\gamma^\mu \hat{n}_{j,\mu}) e^{ik \cdot n_j} \quad (2)$$

where τ is the discrete time step.

Expanding for small k (long wavelength limit), the kinetic term depends on the tensor sum $\mathcal{S}^{\mu\nu} = \sum \hat{n}_j^\mu \hat{n}_j^\nu$. For the FCC lattice, this sum is isotropic:

$$\sum_{j=1}^{12} \hat{n}_j^\mu \hat{n}_j^\nu = 4\delta^{\mu\nu} \quad (3)$$

This yields the effective speed of light:

$$c = 4 \left(\frac{a}{\tau} \right) = 4v_{lattice} \quad (4)$$

This indicates that physical light is a collective excitation moving 4 times faster than the raw lattice updates ($v_{lattice} = a/\tau$). This renormalization factor of 4 is a direct consequence of the FCC coordination number and is a prerequisite for defining the energy scales of the vacuum.

2.2 Geometric Origin of the Planck Scale ($a \approx 0.77l_P$)

With the physical speed of light established ($c = 4a/\tau$), we can now determine the absolute scale of the lattice spacing a . In the SSM, the vacuum energy is distributed across discrete nodes. The maximum energy density the lattice can support is constrained by the volume of the Wigner-Seitz cell. For an FCC lattice with bond length a , the effective volume per node is:

$$V_{node} = \frac{a^3}{\sqrt{2}} \quad (5)$$

The maximum energy a single node can transmit is limited by the lattice update rate τ . Using the derived speed of light relation $c = 4a/\tau$, the maximum energy per mode is:

$$E_{max} = \frac{\hbar}{\tau} = \frac{\hbar c}{4a} \quad (6)$$

The condition for gravitational stability is that the lattice energy density $\rho_{lattice}$ must not exceed the critical Planck energy density ρ_P . We identify the physical vacuum with the maximally packed configuration, where the nodes are strictly compressed against the absolute kinematic exclusion limit of the network, computationally verified in the SSM lattice saturation proofs [3]:

$$r_{min} \geq \frac{1}{\sqrt{3}}L \approx 0.577L \quad (7)$$

At this exact metric wall, the spacing physically cannot become any smaller without violating the fundamental geometric bounds of the tensor network. Consequently, the energy density saturates the gravitational bound but avoids collapsing into an unphysical singularity:

$$\rho_{lattice} = \frac{E_{max}}{V_{node}} = \frac{\hbar c/4a}{a^3/\sqrt{2}} = \frac{\sqrt{2}\hbar c}{4a^4} \quad (8)$$

Setting this equal to the Planck Energy Density $\rho_P = c^7/(\hbar G^2)$:

$$\frac{\sqrt{2}\hbar c}{4a^4} = \frac{c^7}{\hbar G^2} \Rightarrow a^4 = \frac{\sqrt{2}}{4} \frac{\hbar^2 G^2}{c^6} \quad (9)$$

Solving for a :

$$a = \left(\frac{\sqrt{2}}{4} \right)^{1/4} \sqrt{\frac{\hbar G}{c^3}} \approx 0.77l_P \quad (10)$$

Substituting the values of fundamental constants, we obtain a theoretical lattice spacing of $a \approx 1.25 \times 10^{-35}$ m, which is approximately 77% of the conventional Planck length ($l_P \approx 1.62 \times 10^{-35}$ m).

This geometric correction factor arises directly from the specific packing efficiency of the $K = 12$ lattice interacting with the $1/\sqrt{3}L$ metric wall. It demonstrates that the Planck scale is not an arbitrary mathematical regulator, but the exact physical limit where extreme gravitational compression bottoms out against the kinematic exclusion radius of the discrete vacuum.

3 The Continuum Limit and Lorentz Invariance

To validate the model as a physical theory, we must show that it recovers the continuous Dirac equation as $a \rightarrow 0$. We perform a Taylor expansion of the finite difference operator acting on a smooth field $\psi(x)$:

$$\psi(x+n) = \sum_{m=0}^{\infty} \frac{1}{m!} (n \cdot \nabla)^m \psi(x) \quad (11)$$

3.1 First Order (Mass Term)

The $m = 0$ term involves the sum of direction vectors:

$$\sum_{j=1}^{12} \hat{n}_j = 0 \quad (12)$$

This vanishes due to inversion symmetry ($n \rightarrow -n$), ensuring no lattice-induced mass generation.

3.2 Second Order (Kinetic Term)

The $m = 1$ term gives the derivative ∂_μ :

$$\frac{1}{\tau} \sum_j (\gamma \cdot \hat{n}_j) (n_j^\nu \partial_\nu) \psi = \frac{a}{\tau} \gamma^\mu \left(\sum_j \hat{n}_j^\mu \hat{n}_j^\nu \right) \partial_\nu \psi \quad (13)$$

Using the identity $\sum \hat{n}^\mu \hat{n}^\nu = 4\delta^{\mu\nu}$, we recover:

$$4 \left(\frac{a}{\tau} \right) \gamma^\mu \partial_\mu \psi \quad (14)$$

This is the standard relativistic Dirac operator, identifying $c_{phys} = 4(a/\tau)$.

3.3 Third Order (Lorentz Violation)

Crucially, the next order term ($m = 2$, proportional to $a^2 \partial^2$) involves the tensor $\sum n^\mu n^\nu n^\lambda$. Because the lattice is centrosymmetric (invariant under $x \rightarrow -x$), all odd moments of the distribution vanish:

$$\sum_{j=1}^{12} n_j^\mu n_j^\nu n_j^\lambda = 0 \quad (15)$$

This implies that the leading-order Lorentz violating terms are automatically zero. The first deviations appear at $\mathcal{O}(a^2)$, suppressed by the square of the Planck scale. This is consistent with current experimental bounds on Lorentz violation [5, 6].

4 Corollary: Resolution of the Strong CP Problem

The vanishing of odd lattice moments derived in Section III.C has profound implications for the Strong CP problem. The QCD θ -term is given by:

$$\mathcal{L}_\theta = \frac{\theta g^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu} \quad (16)$$

This term violates CP symmetry because the operator $G\tilde{G} \propto E \cdot B$ is a pseudoscalar (odd under Parity). Under the lattice inversion operation $P : x \rightarrow -x$ (which is a symmetry of the FCC vacuum, $P_{latt} = +1$), the pseudoscalar density transforms as:

$$P(G\tilde{G}) = -(G\tilde{G}) \quad (17)$$

Since the vacuum state is invariant under Parity ($P|0\rangle = |0\rangle$), the expectation value of any P-odd operator summed over the centrosymmetric unit cell must vanish:

$$\langle \theta_{QCD} \rangle \propto \sum_{cell} \langle G\tilde{G} \rangle = 0 \quad (18)$$

Thus, the “fine-tuning” $\theta \approx 0$ is actually a geometric identity forced by the O_h point group symmetry of the vacuum. This resolves the Strong CP problem without requiring an axion.

5 Discussion: Obstruction to Graviton Quantization

The discrete nature of the SSM creates a rigid geometric distinction between gauge fields and gravity, specifically driven by the $1/\sqrt{3}L$ metric wall:

- **Fermions (Topological):** Modeled as macroscopic topological braids threading through the lattice links. These are gauge defects characterized by a winding number, which can be strictly localized as discrete quantum states.
- **Gravity (Metric):** Modeled as the macroscopic elastic deformation of the lattice spacing $a(x)$.

This creates an absolute geometric obstruction to quantization. When gravitational compression reaches extreme scales, it does not collapse into an infinitely dense, singular “point particle”. Instead, it strikes the $1/\sqrt{3}L$ kinematic exclusion limit (the metric wall). Because radial compression is strictly halted at this barrier, the immense kinetic energy is forcibly shunted orthogonally into the 2D hexagonal boundary sheets, manifesting as area inflation [7].

Because extreme gravity is inherently a macroscopic tensor deformation bounded by a rigid metric wall rather than a localized, infinitely compressible point defect, a point-like “graviton” is geometrically impossible. This strongly implies that gravity must remain a semi-classical, emergent thermodynamic property of the network in this framework.

6 Conclusion

We have derived the fundamental scales of physics from the geometry of a saturated $K = 12$ vacuum lattice. We showed that the speed of light is a geometric renormalization ($c = 4a/\tau$) arising from the constructive interference of nearest-neighbor paths. Using this relation, we derived the lattice spacing $a \approx 0.77l_P$, demonstrating that the fundamental pixel of the universe is fixed by the packing efficiency of the Cuboctahedron unit cell striking the absolute $1/\sqrt{3}L$ metric wall.

Crucially, the centrosymmetry of the FCC lattice eliminates leading-order Lorentz violations and simultaneously forces the vanishing of the QCD θ parameter, resolving the Strong CP problem. These results suggest that the fundamental symmetries of the Standard Model, and the exact origin of the Planck Scale, are emergent properties of a discrete, centrosymmetric vacuum governed by strict geometric boundaries.

References

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